Dual-Stream YOLOv8 with Cross-Reference Fusion for SMT Component Defect Detection on PCB Production Lines

Vuong Thanh Tung
Faculty of Electronics and Telecommunications
University of Engineering and Technology, VNU
Hanoi, Vietnam

Hoang Van Manh
Faculty of Electronics and Telecommunications
University of Engineering and Technology, VNU
Hanoi, Vietnam

***Abstract*—Automated visual inspection (AVI) of surface-mount-technology (SMT) printed circuit boards is a constrained instance of object detection in which a golden reference image of a known-good board is available at inference time, yet state-of-the-art detectors do not exploit it and operate on the captured frame alone. We propose DS-YOLO (Dual-Stream YOLO), a Siamese extension of YOLOv8s that encodes the capture and the golden through a shared backbone and fuses their multi-scale features (P_3, P_4, P_5) with a learnable Cross-Reference Fusion Module (CRFM). The CRFM adds only $\sim$173 K parameters ($\sim$1.5% of the model) and is identity-initialised, so when the reference is uninformative the network reduces exactly to the YOLOv8s baseline; the gate is self-regulating, enabling line-wide deployment without accuracy regression. Our main contribution is a characterisation of the conditions under which the reference yields a net benefit. For appearance-detectable defects on a single fixed board layout the reference is redundant: DS-YOLO performs on par with the RGB baseline (mAP@0.5$\approx$0.99 for both across 10–50% training-data fractions, mean over three seeds) and the gate remains inactive. On a controlled synthetic benchmark of reference-dependent defects (a component present and locally valid but missing or misplaced relative to the golden), which an appearance-only detector cannot resolve by construction, the gate becomes active and DS-YOLO raises recall from 0.73 to 0.97, confirming that the CRFM exploits the reference when it is informative. The detector runs on a working SMT cell in which a vision server returns OK/NG verdicts to a Mitsubishi MELSEC-Q PLC over OPC UA, and the PLC sorts boards with two articulated robots over CC-Link.**

***Index Terms*—Automated visual inspection, object detection, YOLOv8, dual-stream networks, reference-guided detection, cross-reference fusion, SMT, PCB, OPC UA.**

I. Introduction

Surface-mount-technology (SMT) assembly is the standard process by which modern printed circuit boards (PCBs) are populated. A typical mid-density board carries hundreds of passive and active components placed at very high speed, and even a single missing or misoriented part can render the assembled board non-functional. Visual inspection of every board is therefore mandatory, yet manual inspection is slow, labour-intensive and subject to operator variability. Commercial Automated Optical Inspection (AOI) machines achieve adequate accuracy but incur high acquisition cost and adapt poorly to rapidly changing product mixes, making them ill-suited to low-volume, high-mix electronics manufacturing.

A natural alternative is to combine an industrial camera with a deep object detector trained on the components of interest. The YOLO family of detectors [1], [2] is well suited to this setting owing to its favourable accuracy/throughput trade-off, and several recent works have applied it to PCB defect detection [3]–[5]. A common limitation of these works is that the detector is treated as a generic object detector: it consumes only the captured frame and must internalise, in its weights, the expected appearance of every component at every position. This discards available information, because in an AVI cell a golden reference image of a known-good board is available; it specifies what the line is intended to produce. For a large and practically important class of defects the inspection problem is therefore one of identifying the differences between the capture and the reference rather than localising objects within a single image. This reformulation matters when a defect is reference-dependent, i.e. a component that is present and locally valid but missing or misplaced relative to the golden, for which appearance alone is insufficient to decide OK versus NG. It brings little benefit when defects are already locally visible in the capture; we therefore characterise which regime a given inspection task falls into.

Contributions.

- We propose DS-YOLO, a Siamese dual-stream YOLOv8s with a learnable Cross-Reference Fusion Module (CRFM) that fuses capture and golden features at P_3, P_4, P_5. The CRFM adds only $\sim 173\,\text{K}$ parameters ($\sim 1.5\,\%$ of the model) and is identity-initialised, so its scalar gate is self-regulating: it remains inactive (DS-YOLO $\equiv$ the YOLOv8s baseline, no regression) when the reference is uninformative and becomes active when it is informative, yielding a single model deployable across the line.
- We characterise the conditions under which the golden reference improves detection, on a single SMT board with binary OK/NG labels over four component types (multi-board generalisation is future work). With a unified

protocol and multi-seed runs we show the reference is redundant for appearance-detectable defects (gate $\alpha_\ell \approx 0$, DS-YOLO = RGB), and we construct a controlled synthetic benchmark of reference-dependent defects on which the reference is provably necessary, where the gate becomes active and the CRFM improves detection. This indicates that the module exploits a reference when one is informative, rather than yielding a uniform accuracy gain.
- We integrate the detector into a working SMT inspection cell (Mitsubishi MELSEC-Q PLC, two articulated robots, Basler GigE camera over CC-Link and OPC UA); the requirement to handle reference-dependent defects on future boards motivates the self-regulating design.

The remainder of this paper is organised as follows. Section II reviews related work. Section III formalises reference-guided detection and states our hypotheses. Section IV presents DS-YOLO and the CRFM. Section V summarises the industrial deployment. Experiments are reported in Section VI and Section VII concludes.

II. Related Work

Classical AOI.: Early AOI systems for PCB inspection rely on hand-crafted features such as template matching, normalised cross-correlation and morphological operators [6]. They achieve high precision when the imaging conditions are tightly controlled but degrade rapidly with illumination, color or layout variations and require manual re-tuning for every new board.

Deep detectors for PCB inspection.: Convolutional detectors (Faster R-CNN, SSD and the YOLO family) have largely replaced classical pipelines for component-level defect detection [1]–[4]. These detectors typically treat the inspection task as standard object detection on a single RGB frame and do not leverage the available golden reference. As a consequence, they are prone to over-fitting when the training set is small or class-imbalanced, conditions that are common in real production lines.

Reference- and template-based deep learning.: Several works exploit reference images or template comparison in related domains. Change detection in remote sensing [7] fuses bi-temporal images through Siamese encoders and feature subtraction. Anomaly detection methods such as PaDiM and PatchCore [8], [9] model the distribution of reference patches and flag deviations at inference. These approaches are effective but either require pixel-level labels (change detection) or do not assign semantic classes to the detected anomalies (anomaly detection). Reference-conditioned detection has also been studied directly: one-shot detectors localise a query object by co-attending a support image [10]. The closest line of work for PCB uses reference or template images, e.g. Siamese segmentation of welding defects [11] and template/test image pairs for bare-board copper-trace defects such as open, short and mousebite [5], [12]. These methods target pixel-wise segmentation or trace defects and mostly couple the reference at the input or pixel level. We instead fuse the golden at the feature level through a lightweight, identity-initialised gate. The gate is related to ReZero [13] and LayerScale [14], but it regulates a cross-stream reference difference rather than a residual branch, so a single one-stage detector both localises and classifies SMT component-placement defects and reduces to the appearance-only baseline when the reference is uninformative.

Industrial integration.: On the controller side, CC-Link [15] provides deterministic field-level traffic between a PLC and remote stations, while OPC UA (IEC 62541) [16] offers a vendor-neutral, secure protocol for the PLC-to-supervisory layer. Their combined use is recommended by the RAMI 4.0 reference model and has been reported in several Industry 4.0 case studies [17], [18], but their use as the data plane for a deep-learning vision server has received comparatively little attention.

III. Problem Formulation

Let $\mathbf{I}_g \in \mathbb{R}^{H\times W\times 3}$ denote a fixed RGB image of a known-good (*golden*) PCB, and let $\mathbf{I}_c \in \mathbb{R}^{H\times W\times 3}$ denote a frame captured during inspection. A defect detector is a function

$$\mathcal{F} : \mathcal{X} \longrightarrow \{(b_i, c_i, s_i)\}_{i=1}^{N}, \tag{1}$$

where (b_i, c_i, s_i) is the bounding box, class label and confidence score of the i-th detected component and N is unknown a priori. We distinguish two regimes:

- *Image-only detection* ($\mathcal{X} = \mathbb{R}^{H\times W\times 3}$): the detector observes only the captured frame, $\mathcal{F}(\mathbf{I}_c)$. This is the standard YOLO setting.
- *Reference-guided detection* ($\mathcal{X} = \mathbb{R}^{H\times W\times 3}\times\mathbb{R}^{H\times W\times 3}$): the detector observes both the capture and the reference, $\mathcal{F}(\mathbf{I}_c, \mathbf{I}_g)$.

Reference-guided detection requires the two images to share a geometric frame. On the deployment cell a fixed camera and fixture keep $\mathbf{I}_c$ and $\mathbf{I}_g$ co-registered, so we operate directly on the capture; an optional ORB+RANSAC homography pre-aligns them when this assumption is relaxed (Section IV). DS-YOLO differences features rather than pixels, so it tolerates the small residual drift that remains after registration.

Hypotheses.: We test three hypotheses experimentally:

H1 (defect-dependence) The value of the golden reference depends on the defect type. It is redundant when defects are locally appearance-detectable in the capture (the capture-only baseline already solves the task), but necessary when defects are reference-dependent, i.e. a component present and locally valid yet missing or misplaced relative to the golden, where appearance-only detection cannot decide OK versus NG.

H2 (self-regulating gate) The CRFM's learnable scalar α_ℓ is self-regulating: it remains ≈ 0 (CRFM inactive, so DS-YOLO reduces exactly to the YOLOv8s baseline) when the reference is redundant, and increases when the reference is necessary. The architecture therefore incurs no penalty when the golden is uninformative and exploits it when it is informative.

H3 (no data efficiency) On a single fixed layout a capture-only detector saturates even with limited training data because it can memorise the single fixed layout; the reference does not improve data efficiency in this regime. The benefit of reference fusion is governed by defect type (H1), not by training-set size.

IV. Method: Dual-Stream YOLO

Fig. 1 illustrates the proposed DS-YOLO. The pipeline comprises reference-aware alignment, a shared YOLOv8s backbone applied in Siamese fashion to the capture and the golden reference, and a Cross-Reference Fusion Module (CRFM) that merges the two feature streams before the YOLOv8 detection head.

A. Reference-Aware Alignment

On the production cell the camera and the board fixture are mechanically fixed, so $\mathbf{I}_c$ and $\mathbf{I}_g$ are already co-registered and DS-YOLO operates directly on the capture ($\tilde{\mathbf{I}}_c = \mathbf{I}_c$). When this assumption does not hold, an optional pre-alignment estimates a homography $\mathbf{H}$ from up to K=2000 ORB keypoints (brute-force Hamming matching, Lowe ratio τ=0.75) with RANSAC (σ=3.0 px) and warps the capture into the reference frame. The robustness study in Section VI quantifies tolerance to residual registration error σ_t when this co-registration is imperfect.

B. Cross-Reference Fusion Module

At each scale $\ell \in \{P_3, P_4, P_5\}$ the backbone produces a capture feature map $\mathbf{F}_c^{\,\ell}$ and a golden feature map $\mathbf{F}_g^{\,\ell}$ of equal shape. The Cross-Reference Fusion Module (CRFM) computes a learnable difference attention and modulates the capture features:

$$\mathbf{D}^{\,\ell} = \left|\mathbf{F}_c^{\,\ell} - \mathbf{F}_g^{\,\ell}\right|, \tag{2}$$

$$\mathbf{M}^{\,\ell} = \sigma\big(g_\ell(\mathbf{D}^{\,\ell})\big), \tag{3}$$

$$\mathbf{F}_{\text{fused}}^{\ell} = \mathbf{F}_c^{\,\ell} + \alpha_\ell\, \mathbf{M}^{\,\ell} \odot \mathbf{F}_c^{\,\ell}, \tag{4}$$

where g_ℓ is a 2-layer $1{\times}1$ convolution with a $4\times$ channel bottleneck, σ is the sigmoid, $\odot$ is the element-wise product, and $\alpha_\ell \in \mathbb{R}$ is a learnable scalar initialised to zero. At initialisation $\mathbf{F}_{\text{fused}}^{\ell} = \mathbf{F}_c^{\,\ell}$, so DS-YOLO reduces exactly to a YOLOv8s baseline; as training progresses, α_ℓ and g_ℓ learn the magnitude and spatial distribution of the amplification applied to capture features that deviate from the reference. Because the gate branch is scaled by α_ℓ, the convolutions g_ℓ receive gradient only once α_ℓ departs from zero. This is the mechanism of the self-regulation: the scalar α_ℓ moves off zero only when a consistent reference-dependent error signal is present, so the gate stays at its initialisation (and DS-YOLO at the baseline) when the reference is redundant, and trains only when the reference is necessary, matching the two regimes we observe in Section VI.

C. DS-YOLO Architecture

The full network comprises a single YOLOv8s backbone applied twice (once on $\mathbf{I}_c$ and once on $\mathbf{I}_g$, with shared weights), three CRFM modules at P_3, P_4, P_5, and the unmodified YOLOv8s PANet neck and Detect head. The parameter overhead versus the YOLOv8s baseline is $\sim$173 K parameters ($\sim$ 1.5 % of the 11.3 M-parameter model) concentrated in the three CRFM gates; the compute overhead is one extra backbone forward pass on the golden image, $\sim$50 % FLOPs at inference time. We initialise the backbone, neck and head from the public Ultralytics YOLOv8s pretrained weights [2]; the CRFM modules use Kaiming initialisation for g_ℓ and zero for α_ℓ.

D. Baseline and Loss

We compare DS-YOLO against the single-stream RGB YOLOv8s baseline. The two share the same backbone, the default YOLOv8 loss (distribution-focal + CIoU + binary cross-entropy, no class re-weighting) and the unified training recipe of Section VI, so any difference is attributable to the CRFM rather than to the optimiser or the objective.

V. Industrial Deployment

We integrate the detector into a working SMT inspection cell (Fig. 2). The model deployed in production is the self-regulating DS-YOLO: on the current board it reduces to the RGB baseline (gate $\alpha_\ell \approx 0$, Section VI) at no accuracy cost, and the dual-stream path is retained so the same model will exploit the golden if reference-dependent defects (missing or misplaced parts) appear on future boards; this forward requirement motivates the self-regulating design.

A Mitsubishi Q13UDVCPU PLC with a CC-Link master and a simple-motion module drives the servo and two pick-and-place robots (a Yaskawa GP7 and a Hyundai HH7) over CC-Link.[1] A Basler acA2500 GigE camera feeds a PC-side vision server that subscribes to a capture-trigger signal and writes back the inspection verdict and a completion flag through an OPC UA server (KEPServerEX); the PLC then sorts the board to the OK/NG buffer. In automatic operation the station sustains ≈5.1 s per board (≈81 boards/h), dominated by robot handling, so the extra backbone pass DS-YOLO adds is a small fraction of the cycle.

VI. Experiments

A. Dataset

Images were captured directly on the production cell described in Section V under varying lighting, board placement and component states, all from a single physical board layout. We acquired 75 unique raw captures and applied offline augmentation (flips, HSV jitter, mosaic) to expand the annotated set to 211 images. Each image was annotated manually with component-level bounding boxes over four component types (chip resistor, ceramic capacitor, NE555 timer and CD4017

[1]PLC Q13UDVCPU; CC-Link master QJ61BT11N; motion QD77MS16; servo MR-J4-20B; robot interfaces CCS-PCIE/BD570; HMI Pro-face PFXGP4501TAA.

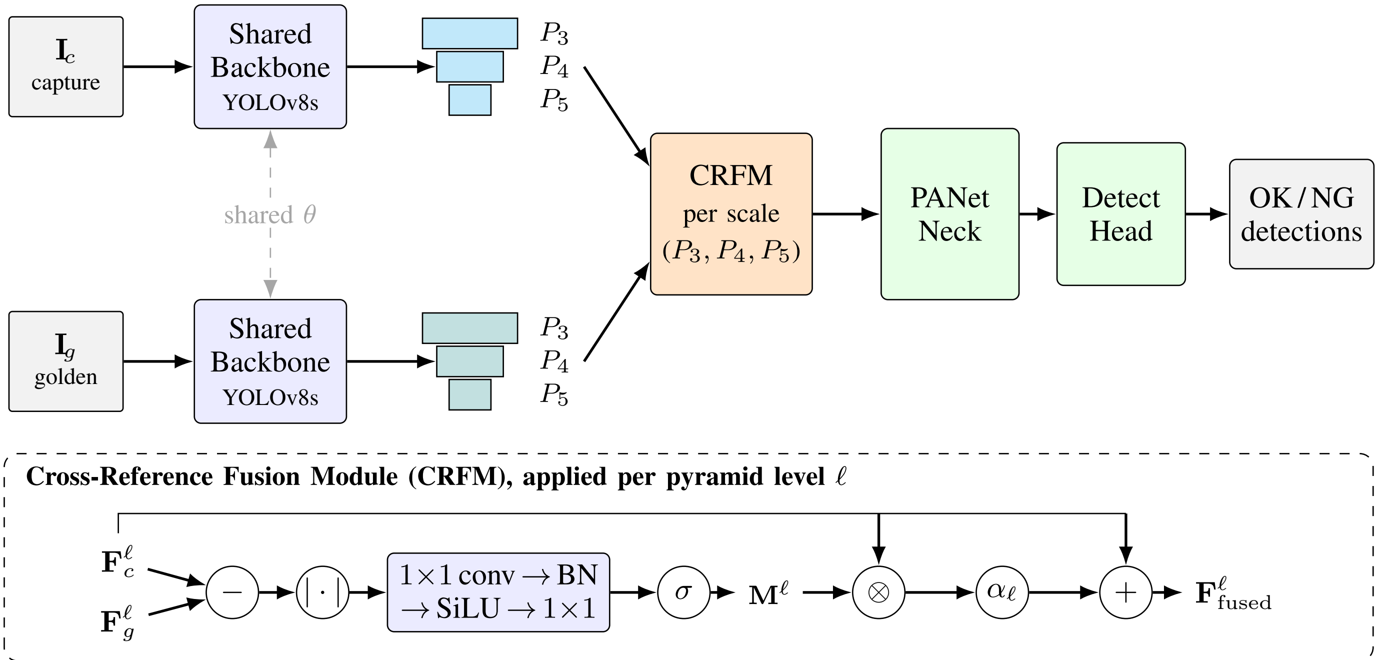

Fig. 1. Architecture of the proposed Dual-Stream YOLO (DS-YOLO). The captured frame $\mathbf{I}_c$ and the always-available golden reference $\mathbf{I}_g$ (pre-registered by the fixed camera and fixture) are encoded by a *shared* (Siamese) YOLOv8s backbone. At every pyramid level $\ell \in \{P_3, P_4, P_5\}$ a Cross-Reference Fusion Module (CRFM) fuses the capture and golden features through a learnable difference-attention gate (inset): $\mathbf{M}^\ell = \sigma\big(g_\ell(|\mathbf{F}_c^\ell - \mathbf{F}_g^\ell|)\big)$ and $\mathbf{F}_{\text{fused}}^\ell = \mathbf{F}_c^\ell + \alpha_\ell\, \mathbf{M}^\ell \odot \mathbf{F}_c^\ell$, with α_ℓ initialised to 0 so the network starts exactly as the YOLOv8s baseline. The fused feature pyramid feeds the unmodified PANet neck and Detect head, which output per-component OK/NG detections.

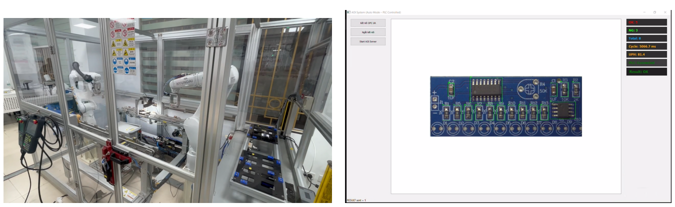

Fig. 2. Deployed SMT cell (left): a Basler GigE camera, a Mitsubishi MELSEC-Q PLC over CC-Link, and two articulated robots (Yaskawa GP7, Hyundai HH7) that sort boards. Operator view (right): per-component OK/NG overlay with measured per-board cycle time and throughput; trigger/verdict are exchanged with the PLC over OPC UA.

decade counter), and each box was assigned a binary status label: OK (the placement matches the design intent) or NG (any defect: missing, shifted or wrongly oriented). The original eight-way labels (per-component OK and NG categories) were collapsed to this binary taxonomy. To avoid data leakage, we split the 75 source captures by source identity (seed 0) into 53 train / 11 val / 11 test sources, then included all augmented copies of each source in the corresponding split (yielding 147 / 11 / 11 images respectively; validation and test keep one un-augmented image per source so the metrics are not inflated by augmentation duplicates). This eliminates the source-level leakage present in the raw export, in which 46 of the 75 sources appeared in more than one split. The resulting val/test sets are small and class-imbalanced by sampling, which we account for when interpreting the metrics. To probe the data-efficiency question (H3) we additionally built training subsets containing 10%, 25% and 50% of the training images, using an identical seed-fixed subset for every variant at each fraction. Statistics are summarised in Table I.

TABLE I
IN-HOUSE DATASET STATISTICS. TRAINING RETAINS EVERY AUGMENTED COPY OF ITS 53 SOURCE CAPTURES (147 IMAGES); VALIDATION AND TEST KEEP ONE UN-AUGMENTED IMAGE PER SOURCE. INSTANCE COUNTS ARE AGGREGATED OVER THE FOUR COMPONENT TYPES.

Property	**Value**
Board layout	1
Unique source captures	75
Augmented pool (Roboflow)	211 images
Source-level split (train / val / test)	53 / 11 / 11
Images used (train / val / test)	147 / 11 / 11
Classes	2 (OK, NG)
Component types	4 (R, C, NE555, CD4017)
Annotated instances (total)	4,314
train (OK / NG)	2,191 / 1,536
val (OK / NG)	85 / 194
test (OK / NG)	217 / 91
OK : NG ratio (overall)	1.37 : 1
Working resolution	640×640

Controlled reference-dependent benchmark.: The in-house defects above are largely appearance-detectable, and a single fixed layout allows a capture-only detector to memorise the correct board. To isolate the regime in which the reference is necessary, we synthesise a reference-dependent benchmark from the same board. Each sample draws a random subset of components to be present (the rest are inpainted out), which defines that sample's layout and its per-sample golden; the capture is the golden with a fraction of the present components corrupted by a missing (inpainted) or shift defect and relabelled NG, while the others remain OK. Because the correct layout varies per sample and is revealed only by the golden, a capture-only detector cannot distinguish a missing-NG slot from a legitimately absent one and must therefore use the reference. We generate $\sim$1,500 samples, split 70/15/15 by source capture so that no raw board image appears in more than one of train, validation and test. This circularity is intentional: the benchmark defines a task that a capture-only model cannot solve, allowing us to test whether the gated reference path recovers the otherwise-unattainable performance. The benchmark therefore probes the fusion mechanism rather than evaluating realistic field performance; validating the effect on physically-induced reference-dependent defects is left to future work.

B. Implementation Details

Both models were trained for 100 epochs at input size 640×640 with batch size 4 on an NVIDIA Tesla T4 GPU (15 GB) using SGD with lr_0=0.01, momentum=0.937, weight decay=5×10^{-4} and a cosine schedule with 3-epoch warmup. The RGB baseline uses the Ultralytics pipeline; DS-YOLO uses a custom PyTorch loop, since the dual-stream forward pass requires injecting the golden image. To keep the comparison fair, both share the same optimiser, schedule, augmentation and best-checkpoint criterion (validation mAP@0.5), and both are scored through identical post-processing (multi-label NMS, conf 0.001, IoU 0.7, max-det 300); a cross-check re-scores the same checkpoint through both code paths and requires the two mAP columns to agree within 0.002. Both are initialised from the public YOLOv8s pretrained weights; the CRFM convolutions use Kaiming weights and $\alpha_\ell = 0$ (identity at initialisation, with BatchNorm frozen on the golden pass so that at $\alpha_\ell = 0$ the network is architecturally identical to the single-stream YOLOv8s baseline). Augmentation was limited to HSV jitter and horizontal flip; mosaic and strong geometric augmentations were disabled, as they would break the capture–golden alignment that reference-guided fusion relies on. The in-house data-fraction results are reported over three seeds as mean±std; the synthetic benchmark is a single run, with each capture loading its own per-sample golden by filename.

C. Main Results

The value of the golden reference depends on the defect type, and the two regimes yield contrasting results.

Regime 1: appearance-detectable defects (reference redundant).: On the in-house board the capture-only RGB baseline already saturates and DS-YOLO performs on par with it (Table II). Across 10–50% training-data fractions (mean±std over three seeds) the two are equal within run-to-run variance at mAP@0.5≈0.99, and on the more demanding mAP@0.5:0.95 the gap is likewise within seed noise. The fusion gate is inactive throughout ($|\alpha_\ell| < 10^{-3}$ at convergence for all three scales), so DS-YOLO is equivalent to the YOLOv8s baseline and incurs no accuracy regression. Under inference-time translation noise up to 20 px, with labels warped along with the image, both detectors remain flat in $[0.991, 0.993]$; since $\alpha_\ell \approx 0$ this does not probe the fusion. The equivalence is a non-regression result on a small test set ($n = 11$ images, 308 instances), not a powered equivalence test.

TABLE II
REGIME 1 (IN-HOUSE BOARD, REFERENCE REDUNDANT): RGB VS. DS-YOLO, MEAN±STD OVER THREE SEEDS. THE TWO COINCIDE WITHIN SEED NOISE AND THE FUSION GATE STAYS $\alpha_\ell \approx 0$.

In-house setting	**mAP@0.5**		**mAP@0.5:0.95**	
	RGB	DS-YOLO	RGB	DS-YOLO
10% train data	0.991±.002	0.989±.003	0.813±.008	0.805±.008
25% train data	0.986±.003	0.983±.008	0.821±.003	0.814±.007
50% train data	0.991±.001	0.986±.006	0.827±.009	0.815±.006

TABLE III
REGIME 2 (SYNTHETIC REFERENCE-DEPENDENT BENCHMARK): THE TASK IS BY CONSTRUCTION UNSOLVABLE FOR A CAPTURE-ONLY MODEL, SO THE RGB ROW IS ITS DESIGN CEILING, NOT A TUNED COMPETITOR. THE EXPERIMENT ASKS ONLY WHETHER DS-YOLO'S GATE BECOMES ACTIVE AND IMPROVES DETECTION. SINGLE RUN ON $\sim$1,500 SAMPLES.

Method	**Prec.**	**Recall**	**mAP@0.5**	**mAP@0.5:0.95**
YOLOv8s (RGB, capture only)	0.925	0.733	0.855	0.721
DS-YOLO (ours)	**0.990**	**0.971**	**0.992**	**0.863**

Regime 2: reference-dependent defects (the reference probe).: By construction the synthetic benchmark is unsolvable for a capture-only model: a missing-NG slot and a legitimately absent slot are pixel-identical in the capture, so the recall of the RGB baseline is capped before any training. The experiment is therefore not a performance comparison but a test of whether the gate becomes active and the CRFM exploits a reference it provably requires (Table III). The RGB baseline plateaus at mAP@0.5$=$ 0.85, recall $=$ 0.73 (its by-design ceiling; the residual recall comes from the appearance-visible shift defects). DS-YOLO's gate becomes active (α_ℓ grows to -0.19, -0.67 and -0.01 at P_3, P_4, P_5, against $|\alpha_\ell| < 10^{-3}$ on the in-house board) and the model improves detection to mAP@0.5$=$ 0.99, recall $=$ 0.97, with the gain concentrated on the missing components that carry no local cue. Since at $\alpha_\ell = 0$ DS-YOLO's forward pass is architecturally identical to single-stream YOLOv8s (BatchNorm frozen on the golden pass), the architecture cannot exceed the capture-only baseline other than through the CRFM, so the gate activation indicates that the module uses the reference.

Caveats. This reference-dependent result is synthetic and from a single run: the defects are inpainted, the RGB ceiling is fixed by the task design, and we do not estimate variance. Its purpose is to verify that the gate can exploit a known-necessary reference rather than to estimate field accuracy. Validating the effect on physically-induced reference-dependent defects, over multiple seeds and with an inpainting-artefact control, is the most important next step.

D. Data Efficiency (H3)

We trained both models on 10%, 25% and 50% of the in-house set, using an identical seed-fixed subset at each fraction and averaging over three seeds (Table II). Although one might expect a reference to help most when data is scarce, the two models are equal within run-to-run variance at every fraction, including 10% (≈ 15 images): a single fixed layout is learned from few captures, so the reference adds no measurable benefit. Reference fusion therefore improves defect coverage (H1) rather than data efficiency (H3). We note that an earlier version of this study reported a low-data advantage that was later found to be a training-subset artefact.

E. Latency

The dual-stream forward adds a second backbone pass on the golden ($\approx 1.5\times$ the RGB inference cost); at deploy time the golden features are cached once per board type, leaving only the CRFM gates and the extra neck input. Against the robot-dominated ≈ 5.1 s per-board cycle (Section V), this overhead is negligible.

F. Discussion and Limitations

Our results support all three hypotheses: the value of reference fusion is conditional on the defect type. Its benefit is governed by defect type (H1), large for reference-dependent defects and negligible for appearance-detectable ones, and the self-regulating gate (H2) makes DS-YOLO safe to deploy even when the reference is uninformative, since it then reduces exactly to the baseline. The main limitation is that the reference-dependent gains are demonstrated on a synthetic benchmark; although it is constructed so the golden is provably necessary, confirming the effect on physically-induced defects (a board deliberately populated with misplacements) is the most important next step. Two further limitations: DS-YOLO assumes the golden is registrable to the capture (met here by a fixed camera and fixture), and like any supervised detector it can only flag labelled defect modes; combining it with an unsupervised anomaly branch (e.g. PaDiM [8]) is a natural extension.

VII. Conclusion

We have presented Dual-Stream YOLO (DS-YOLO), a Siamese extension of YOLOv8s that fuses the always-available golden reference into the detector via a lightweight (~ 173 K-parameter), identity-initialised Cross-Reference Fusion Module. Rather than claim a uniform improvement, we characterise the conditions under which the reference provides a measurable benefit. On appearance-detectable defects from a single fixed board the reference is redundant and DS-YOLO performs on par with the RGB baseline (the fusion gate stays ≈ 0); on reference-dependent defects (a component present but missing or misplaced relative to the golden) it is necessary, and DS-YOLO improves recall by $+24$ points (mAP@0.5 $0.85 \rightarrow 0.99$) over an appearance-only detector, with the gate active. The system is deployed on a real SMT cell integrating a Mitsubishi PLC, two articulated robots and a GigE camera over CC-Link and OPC UA, adding only one extra backbone pass per board. Future work will validate the reference-dependent gain on physically-induced defects, extend the study to multiple board layouts, and integrate an unsupervised anomaly branch for unlabelled defect modes.

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